

SINGLE PHASE INDUCTION MOTORS.

1

Electric current flowing through a wire cause magnetic flux around the ~~the~~ conductor.

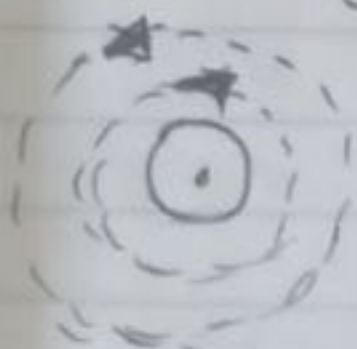
direction of the magnetic flux depends on the direction of the current.

If we draw a small circle to represent a conductor, a current through this conductor can either be entering or leaving the paper on which the conductor is drawn.

A current be represented by a dot if it is leaving the paper and a plus sign $+$ if it is entering the paper. i.e.

⊙ leaving paper ⊕ entering paper.

Using the right hand grip rule which says if you grip a conductor carrying current with your right hand and let the thumb point in the direction of the current, then the fingers will point in the direction of the magnetic flux.



anti clockwise



clockwise

2
If a conductor is placed in moving magnetic flux
emf is induced in the conductor.

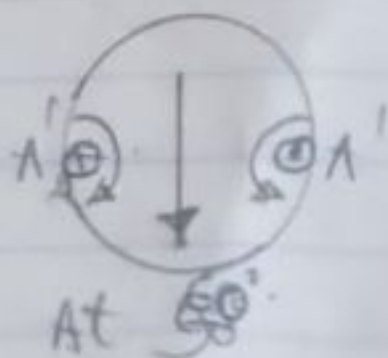
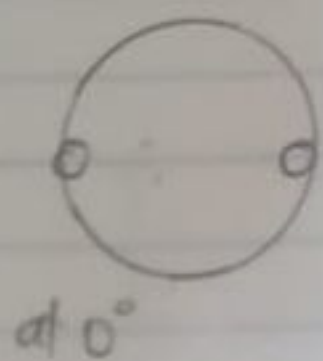
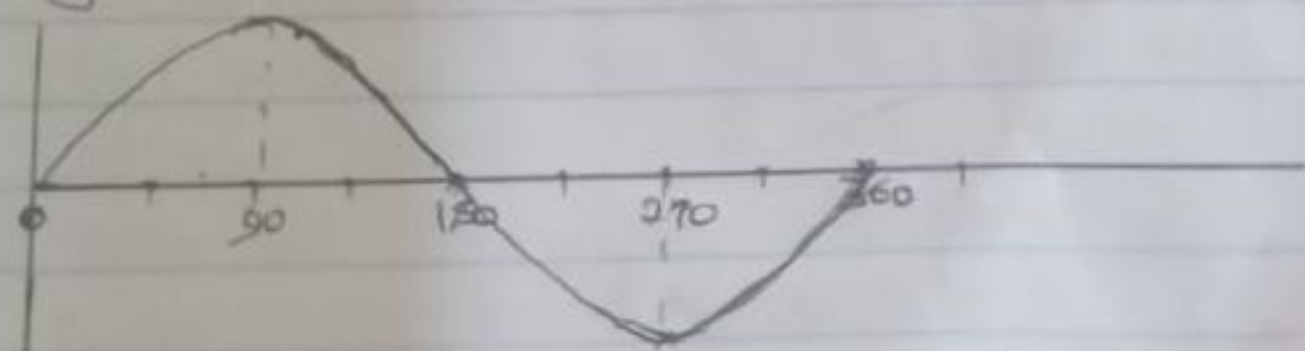
Further, if a conductor is moved in a
magnetic field emf is induced in the conductor.

This process of generating emf in a conductor is
called induction, because emf is induced in
the conductor.

Induction motors have stator winding for the purpose of
magnetic flux production and rotor for
magnetic flux link with the conductors of the rotor.
emf is induced in the rotor.

Single phase induction motors are not self starting, be-
cause they produce a pulsating flux. The pulsating
flux act along 180° producing zero net torque.

pulsating current in the stator





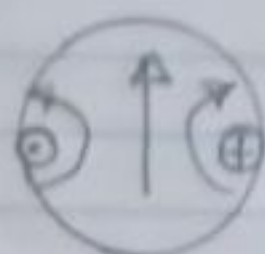
At 120°



At 150°



At 180°



At 240°



At 300°

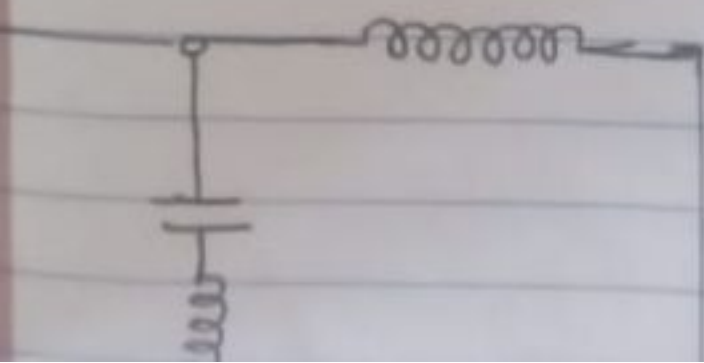


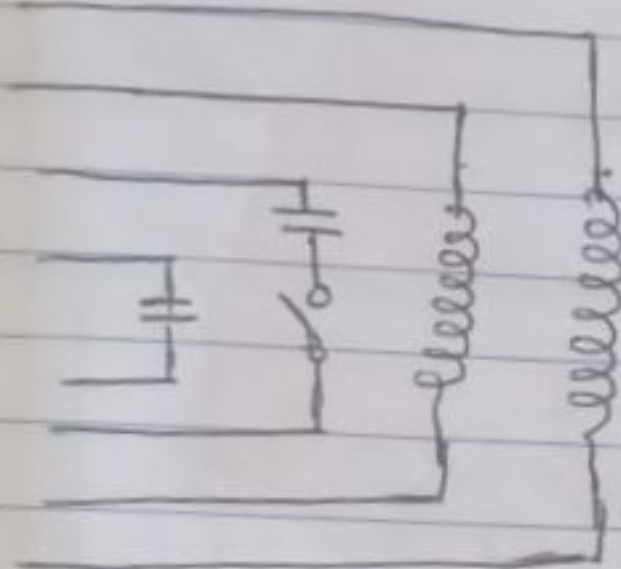
At 360°

Single winding in single phase motor produce a pulsating flux. The pulsating flux act along a single line with zero net torque.

In single winding the motor hums and does not rotate. motor over heats quickly and therefore should not left connected for long because it burns the wdg. never if the motor is given a kick it runs in direction of the kick.

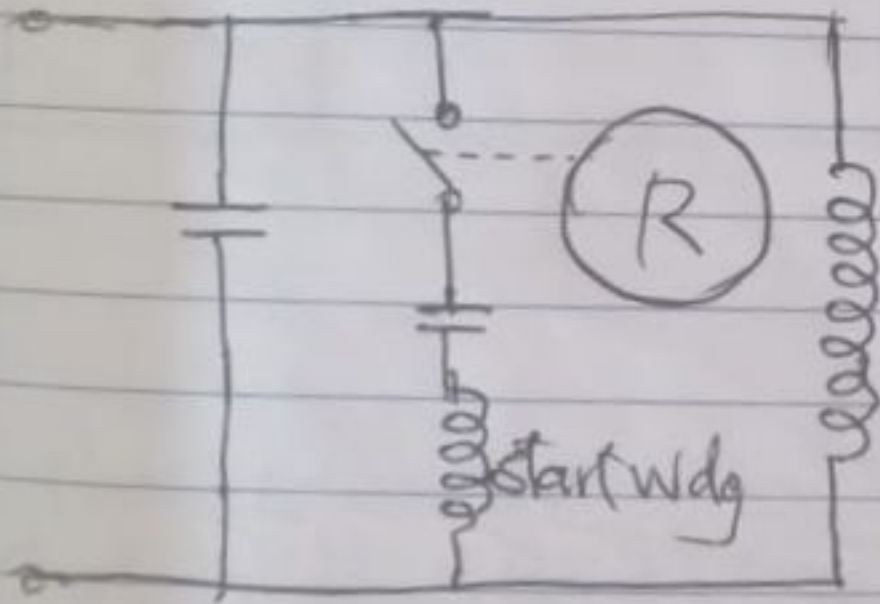
to make single phase induction motor self starting we introduce another wdg, the start wdg. start wdg is connected in series with a capacitor. capacitor makes the wdg more capacitive than the wdg. The two wdgs are connected in parallel. current through the capacitive (start wdg) is less than the current through the run wdg.





All components of the single phase induction motor are connected at the terminal block of the motor.

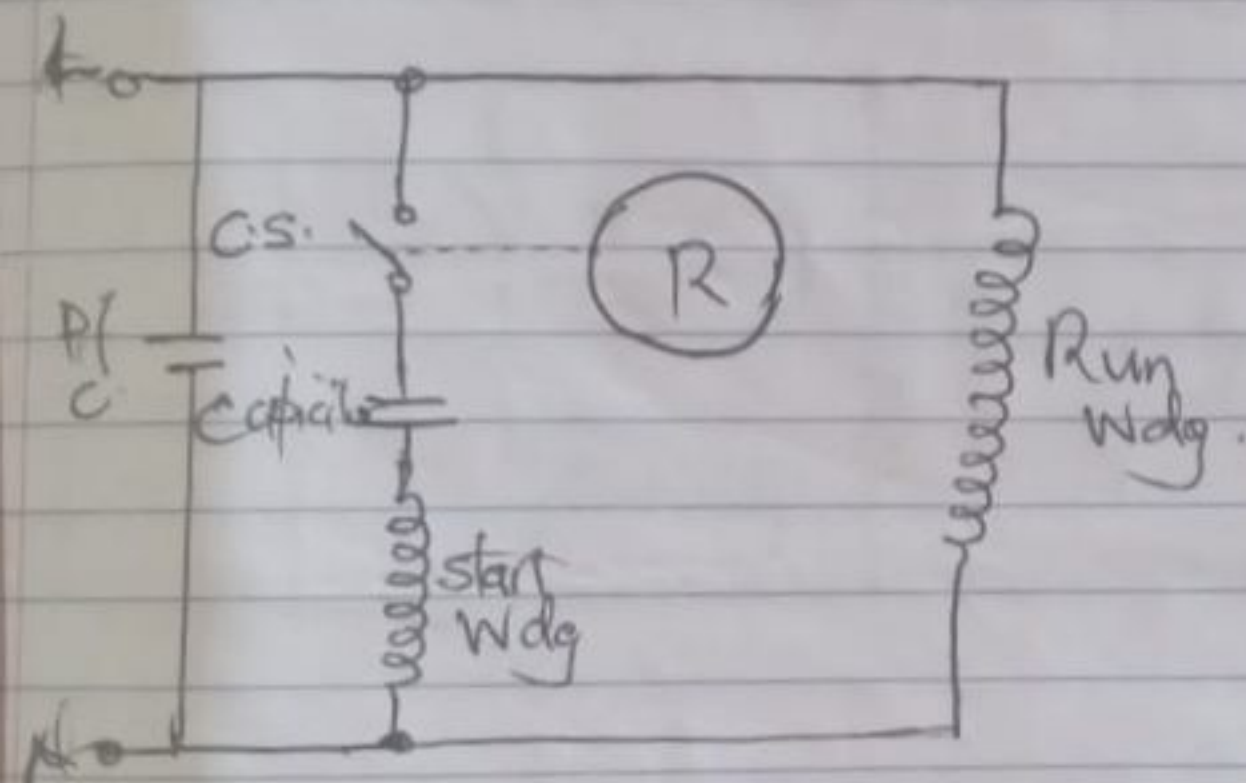
At the terminal block the components are connected to make an internal wiring as below.



Run Wdg.

The flux due to run and starting wdg rotates.
This rotating flux produce starting torque, making the motor self starting.

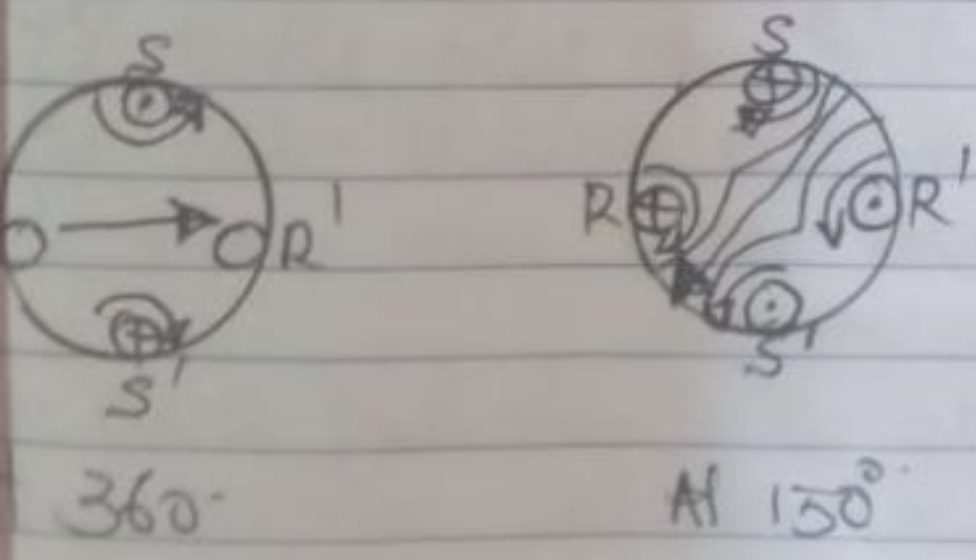
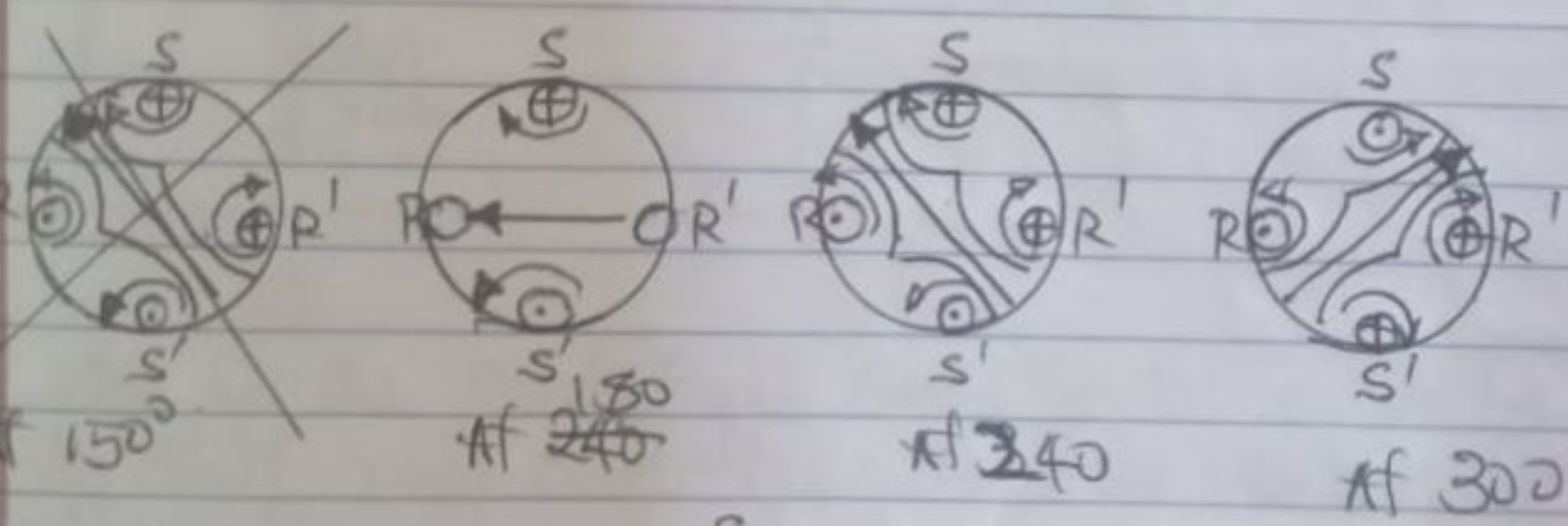
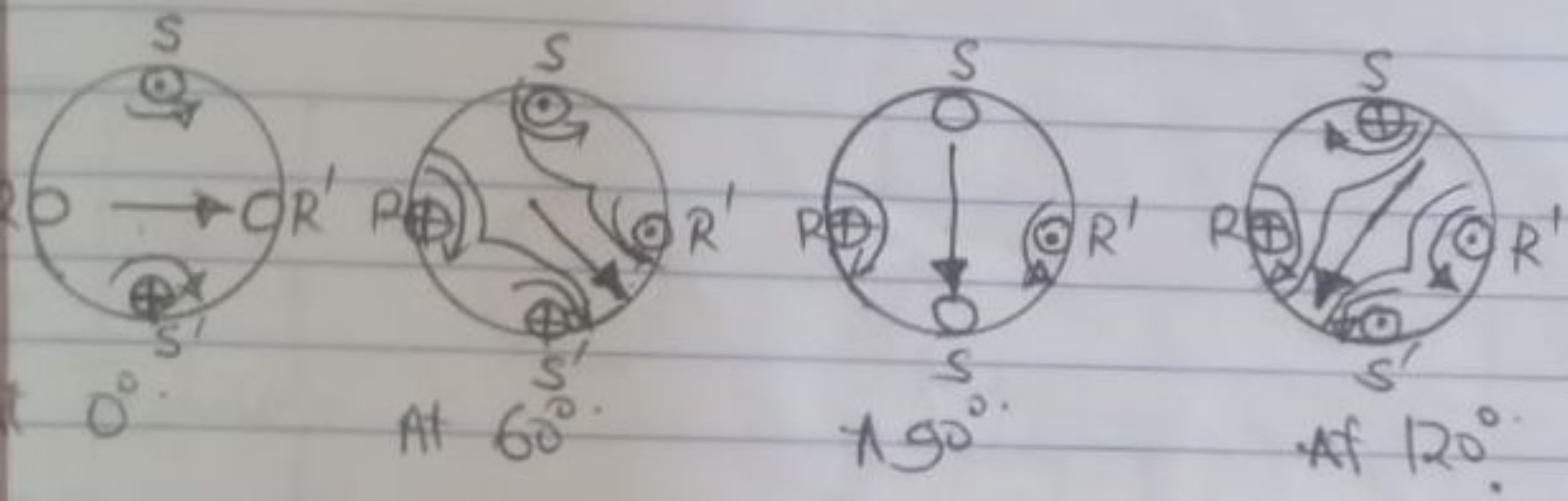
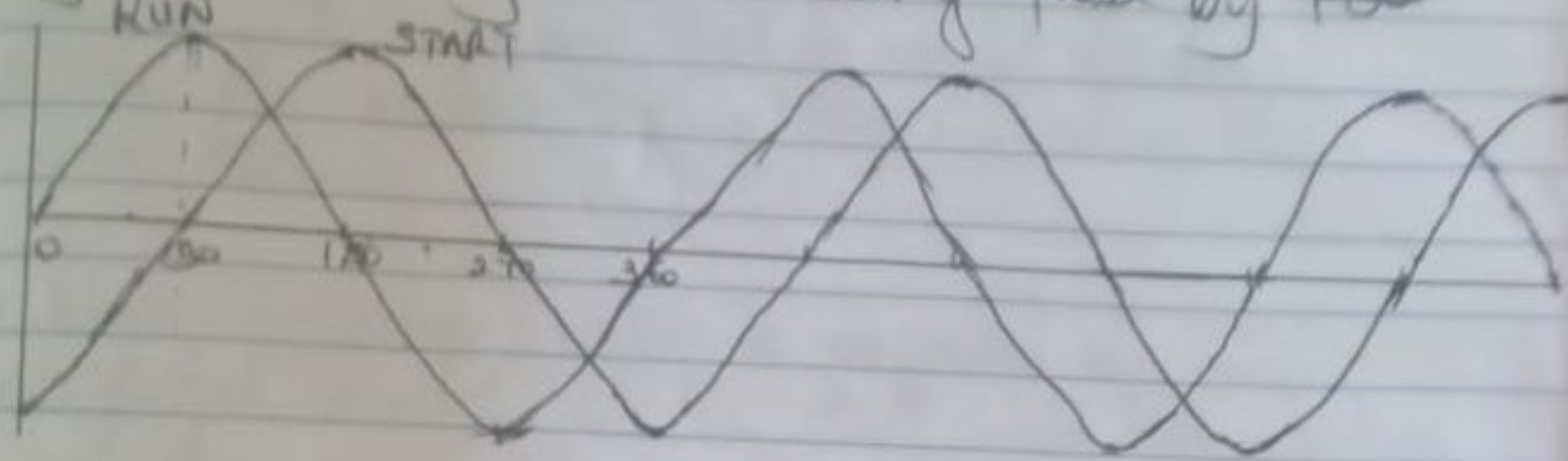
Single phase induction motor is improved by adding a centrifugal switch in the start wdg branch, as seen below.



The role of the centrifugal switch is to isolate (disconnect) the start wdg once the motor has started; at 67% of the final speeds. This improves the efficiency of the motor because retaining the starting wdg in c. would mean more power loss.

Single phase induction motor is further improved by adding another capacitor for improving power factor.

Assuming the two waves will be out of phase by 90°



10
12

motor is started because the device will not allow the starting current to flow to the motor.

Overload protective devices for motors are fitted with a reset button, so that the device can be reset after it has isolated the motor. This is done manually by the technician attending to the motor that has been isolated ~~by~~ due to overload.

Before resetting the protective device the technician should first try and establish the cause of the overload and deal with it before resetting the device.

Overload current is sensed in the power circuit at the overload device but it is operated in the control circuit.

^{induction}
When starting, motors draw 8-10 times their running current as the starting current. For this reason motors should not be started on load, otherwise the ~~motor~~ motor will draw excess current that can be detrimental to the ~~wires~~ windings. Even under no load motors will draw excess current when starting. This excess current that occurs during starting is classified among transient faults. Transient faults are faults ^{that} clear on themselves within a short period and they cause no damage.

Protective devices should accommodate transient faults. They should allow them to clear without operating. To accommodate transient faults, protective devices

The overload device is usually connected in a position to intercept excess current before it reaches the motor i.e. before the motor and not after the motor.

Overload devices for motors are variable i.e. they protect current in a given range. They should be set to match the load current the particular motor is likely to draw while in operation. The technician doing motor installation should calculate the load current from the rating given in the name plate of the motor.

Example if a motor is rated 3HP, this should be converted to Watts

$$3 \times 746 = 2238 \text{ W} \quad (1 \text{ HP} = 746 \text{ W})$$

from formulae $P = VI \cos \phi$

$$I = \frac{P}{V \cos \phi}$$

$$I = \frac{2238}{415 \times 0.86} = 6.27 \text{ A}$$

where
0.86 is
the PF
of the motor

The setting of the overload protective device for this particular motor should be set at 6.3A

NB If the range of protection for the device is up to 30A, the device will not protect the motor if set at a Maximum.

On the other hand if the setting on the protective device is on the lower range say at 3 or 2A for this motor, this will cause a problem of nuisance tripping whenever the

Motor protection

Motor protection (i) Overload.

Motors are protected against overload.

Overload in motors can be caused by

- (i) Short circuit between live and neutral
- (ii) " " " " live and earth
- (iii) " " " " live and live.
- (iv) excessive mechanical load causing motor to draw more current.
- (v) Motor starting (vi) Motor Stalling.

Effects of overload in motors can be short term or long term. Short term effect can be burning of the windings while the long term effect could be deterioration of the insulation due to heating effect of the excess current drawn by the motor.

Motors are protected against overload by overload protective device. The overload protective device can be thermal or magnetic. This device is operated to isolate the motor whenever excess current flows as per the setting of the device. The safe / normal current cannot cause the device to operate since the heat generated is below the set value for the thermal overload. In the case of the magnetic overload, the magnetic flux caused by the motor ^{current} has magnetomotive force in the lower range that cannot operate the device. However in the event that the motor draws excess current, this current generated enough heat to operate thermal overload or sufficient mf to operate the device.

have time lag that make them to drag in operating, thus allowing the transient fault to clear.

Self starting

Motors should be protected against self starting, i.e. in the event that power fails when motors are in operation they should not re-start automatically when power is restored.

This is made possible by use of magnetic contactor switches in the power circuits and mechanical switches (push buttons) in the control circuits. Motor contactor switches are operated by passing current through the contactor coil. The current through the contactor coil is controlled by use of mechanical push buttons in the control circuit.

Motor load rating / Motor rated power output.
This is numerical value of electrical KW or mechanical HP that signify the ^{max.} demand made on electrical circuit at a given instant by a motor, and is proportional to the load connected.

2) Motor no-load

This ~~refer~~ refer to the state of a machine (motor) rotating with zero output power under normal operating conditions.

3) Motor full load

Thus the load which causes motor to operate at its rating. (KW or HP)

4) Duty of a motor.

This is a continuous, short time or periodic duty comprising one or more loads remaining constant for the condition specified, or a non periodic duty in which generally load and speed varying with the permissible operating range.

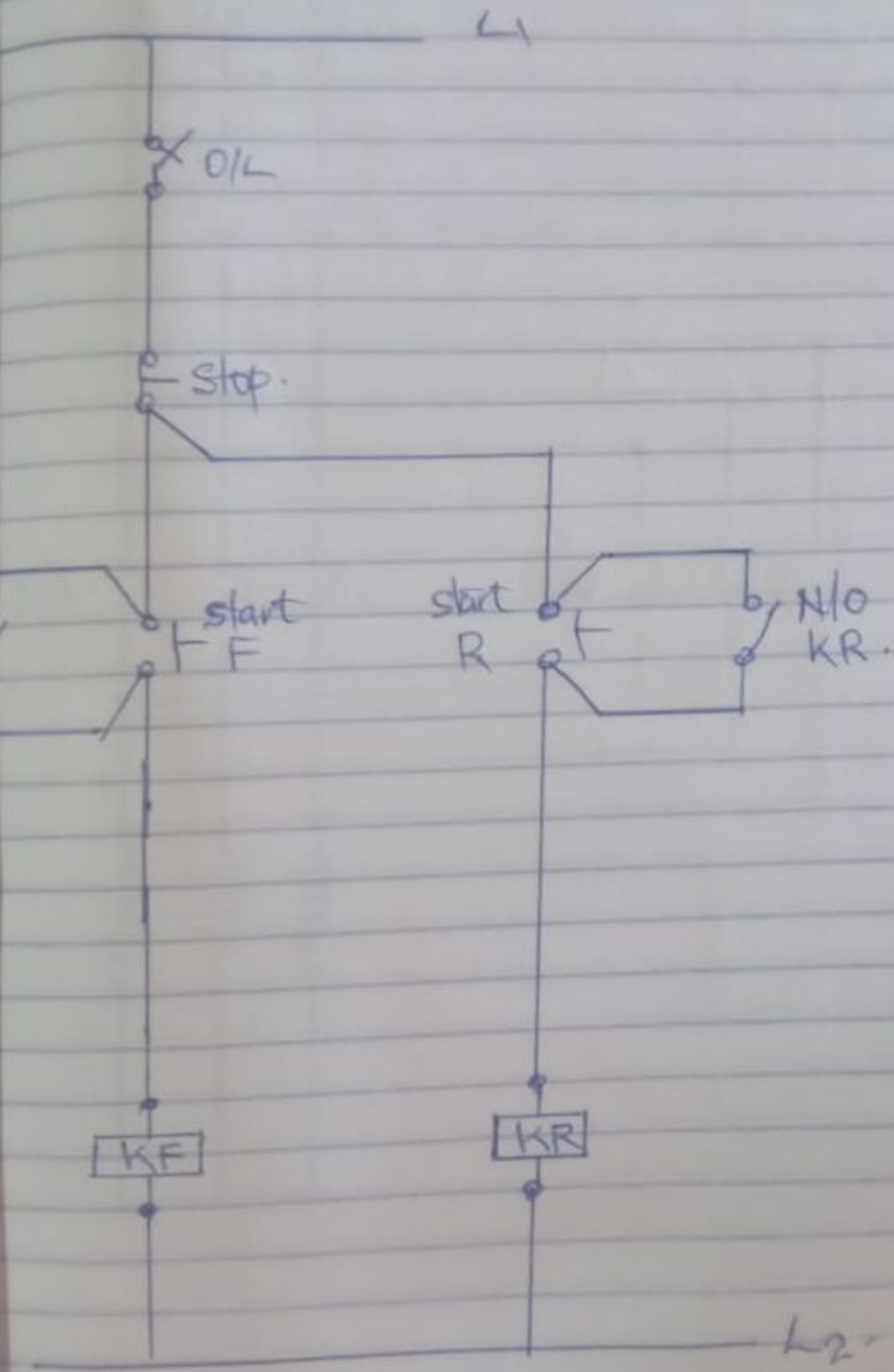
5) Locked Rotor Torque.

This is the smallest steady-state rms current taken from the line with the motor held at rest over all angular positions of its rotor at rated voltage and frequency.

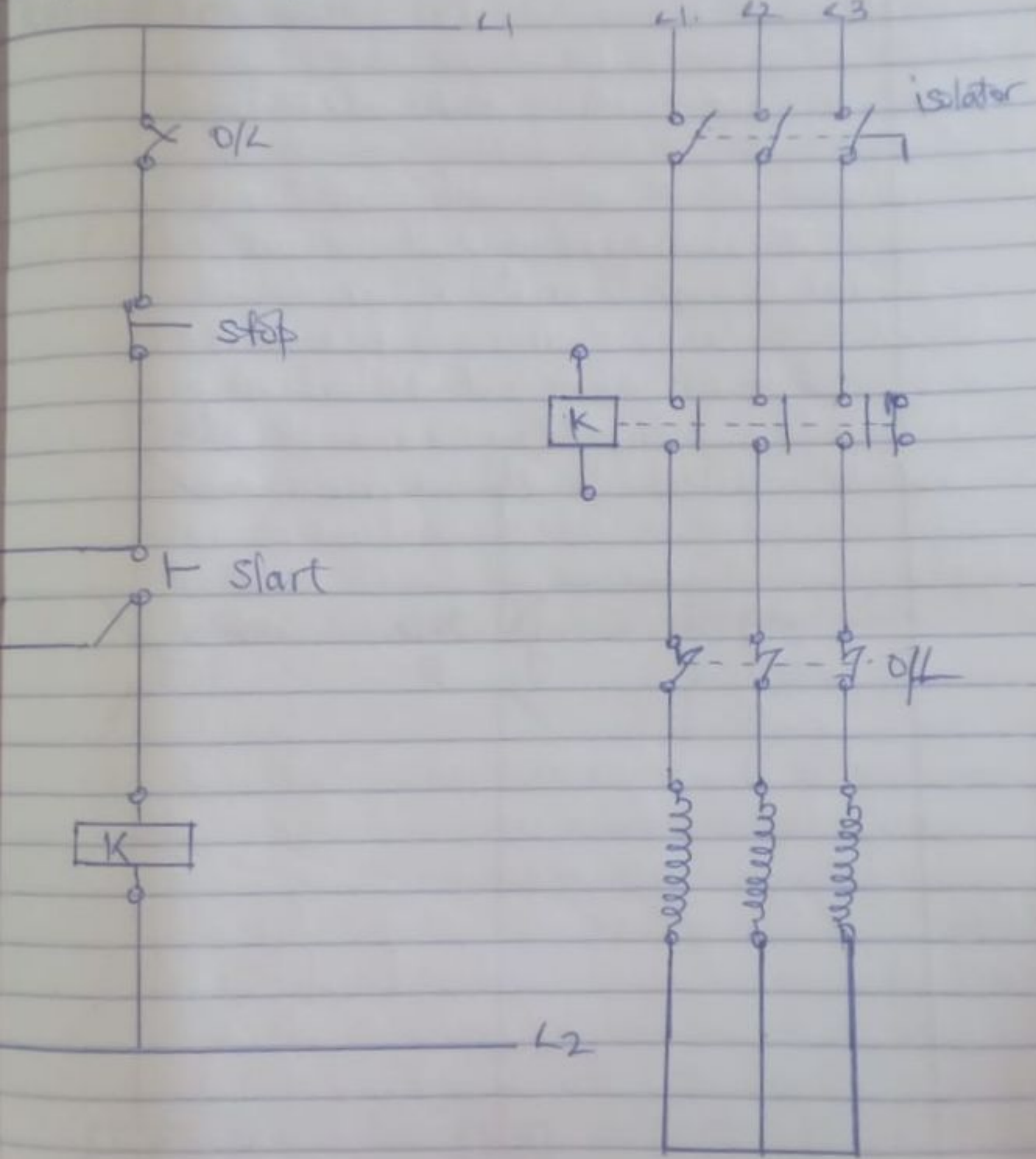
- (6) **Pull-up torque**
For AC motors, is the smallest steady state asynchronous torque, which the motor develops between zero speed and the speed which corresponds to the break-down torque when the motor is supplied at the rated voltage and frequency.
- (7) **Breakdown Torque**
For AC Motors, is the maximum steady state asynchronous torque which the motor develops without an abrupt drop in speed, when the motor is supplied at the rated voltage and frequency.
- (8) **Cooling**
This is a ~~procedure~~ by means ~~of which~~ ^{by which} heat resulting from losses occurring in a machine is given up to a primary coolant which may be continuously replaced or may itself be cooled by a secondary coolant in heat exchanger.
- (9) **Coolant**
This is a medium either liquid or gas by which heat is transferred.
- (10) **Thermal equilibrium**
This is the state when temperature gain equals temperature loss.

3Φ DOL FORWARD-REVERSE OPERATION.

control circuit



3 ϕ DOL FORWARD STARTER OPERATION



(11) Motor efficiency
This is the ratio of output power to input power expressed in the same units and given as a percentage.

(12) No-load Test
For motors, this is performed to establish motor demand from a circuit when yielding no mechanical output. For generators, this is performed to establish generator demand from its prime mover while its terminals are open circuited.

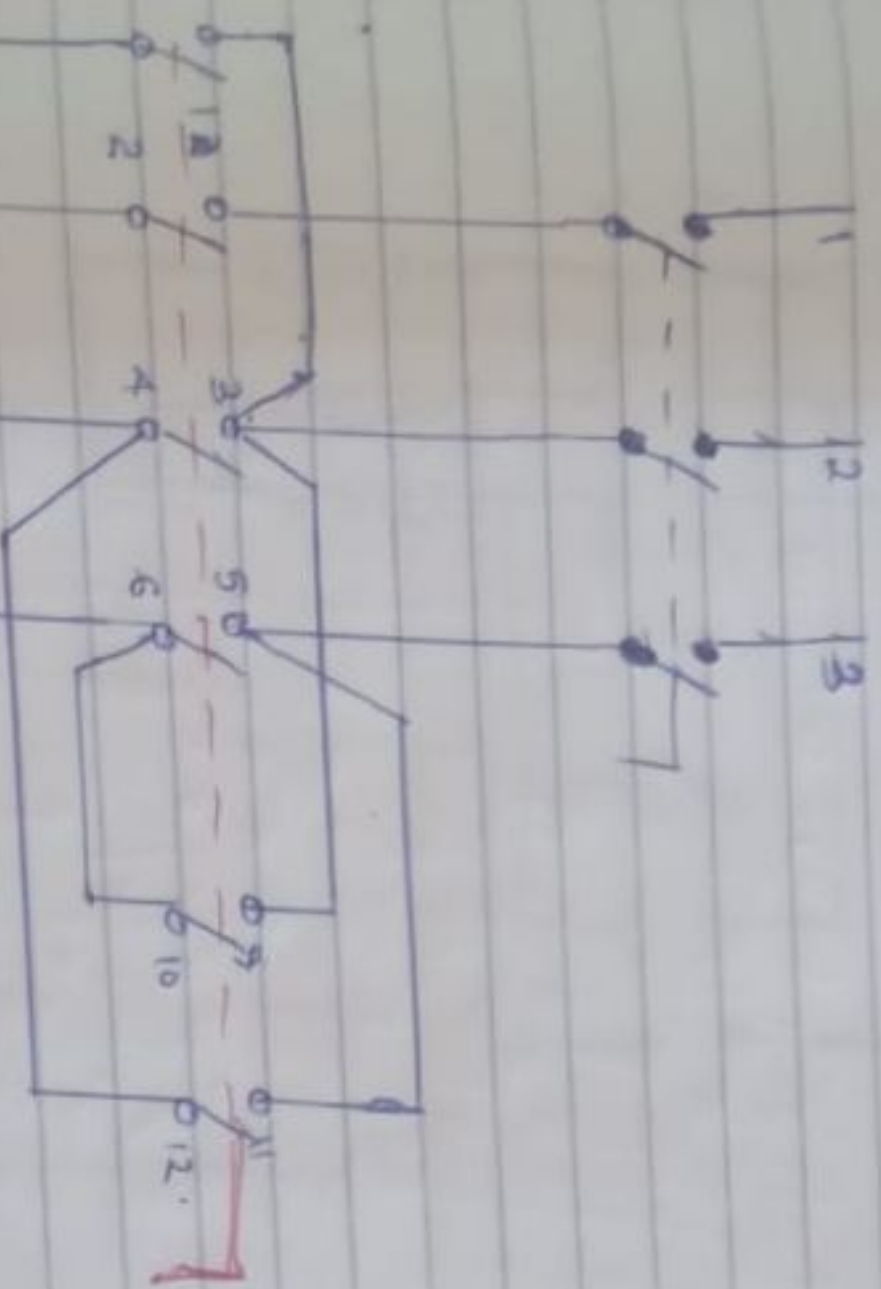
(13) Locked rotor Test

(14) Motor total losses
This is the difference between the input and the output power, equivalent to the sum of the constant losses, load losses, additional load losses and the excitation losses.

(15) Constant losses
This is the sum of the iron losses, and the friction and windage losses.

1 3 5 7 9 11
2 4 6 8 10 12

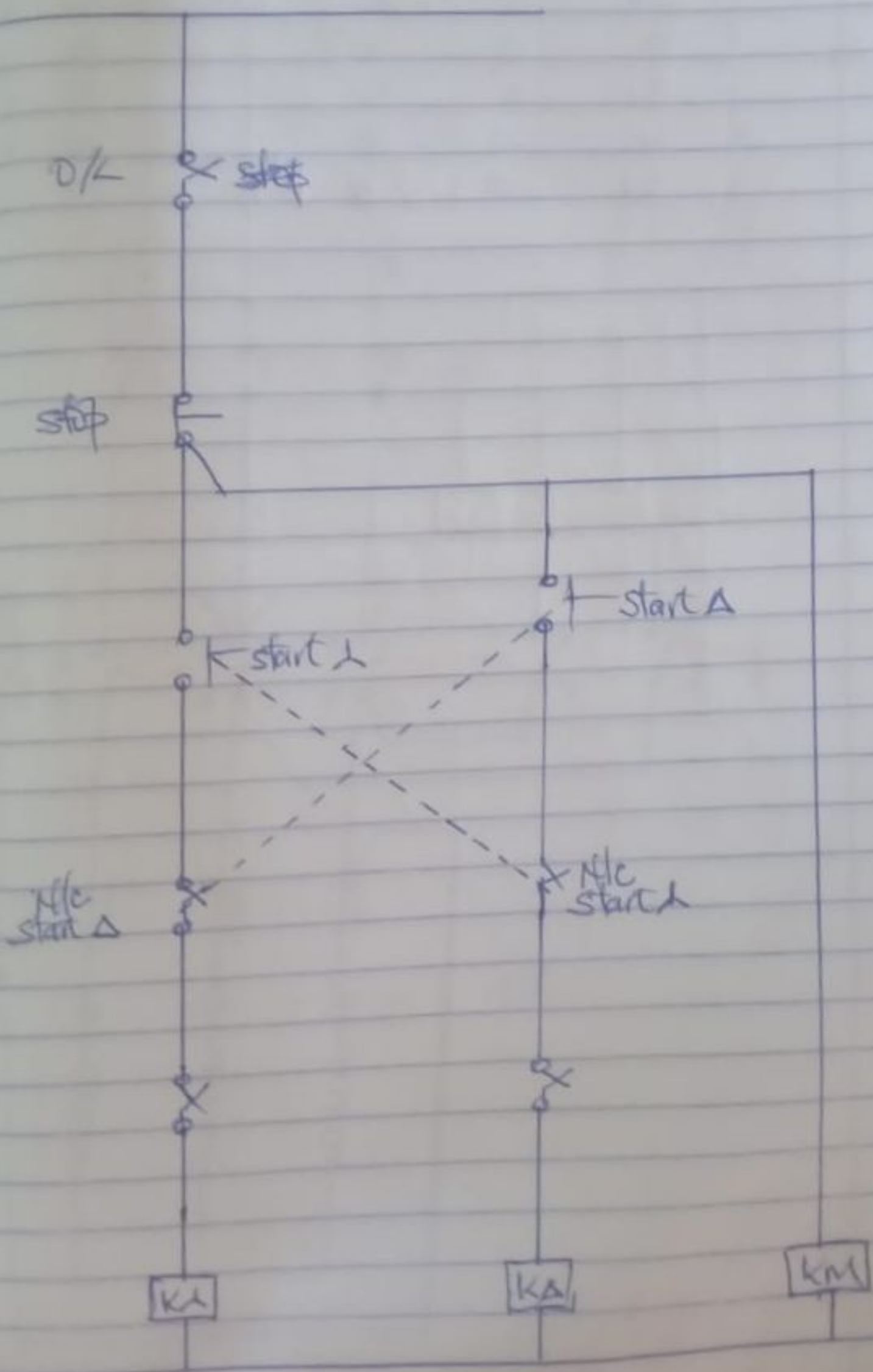
	1	0	2	
1	✓	0	✓	}
2	✓	0	0	
3	✓	0	0	
4	✓	0	✓	}
5	✓	0	✓	
6	0	0	✓	



-O/K.

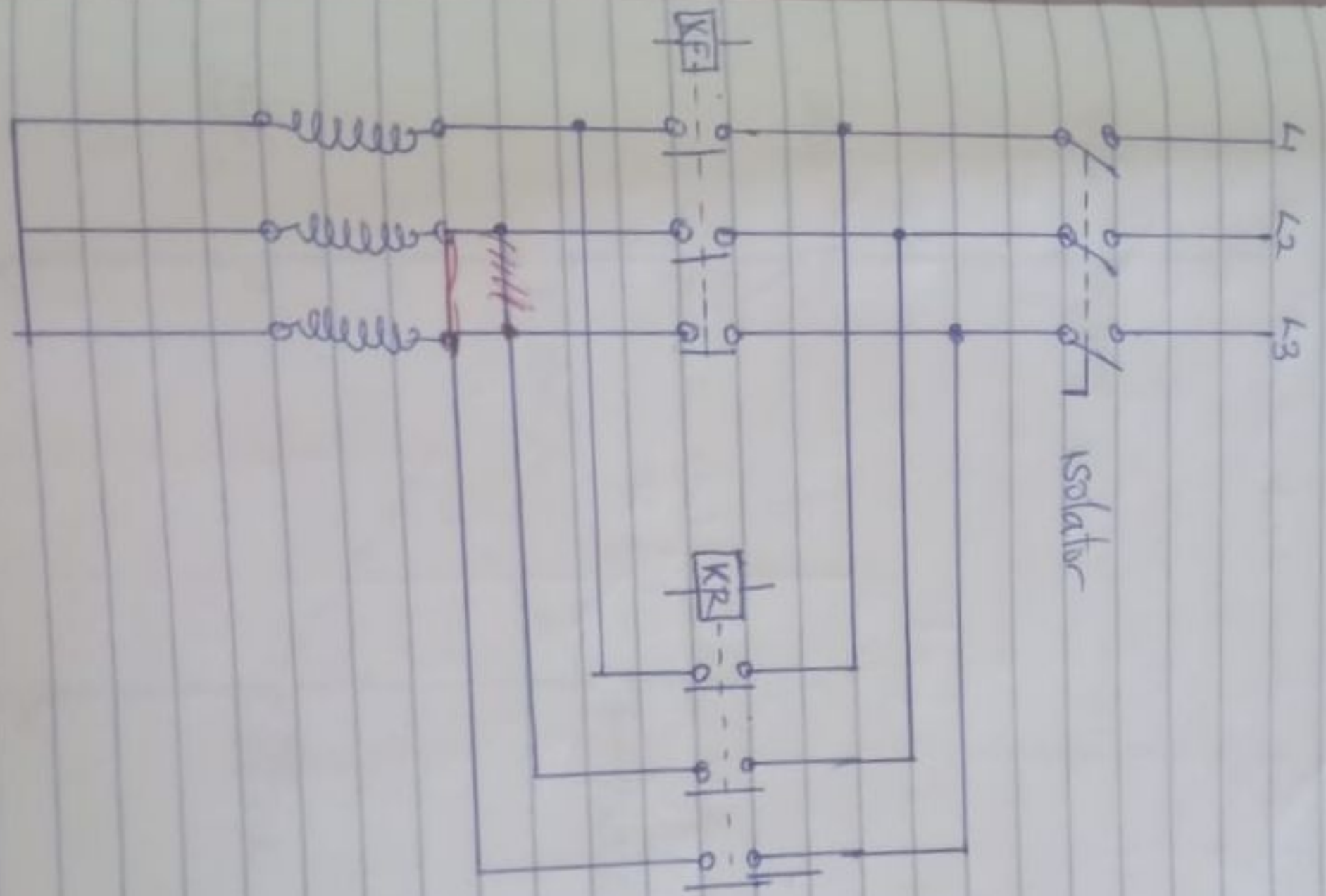
3φ STAR-DELTA OPERATION (MANUAL)

control circuit

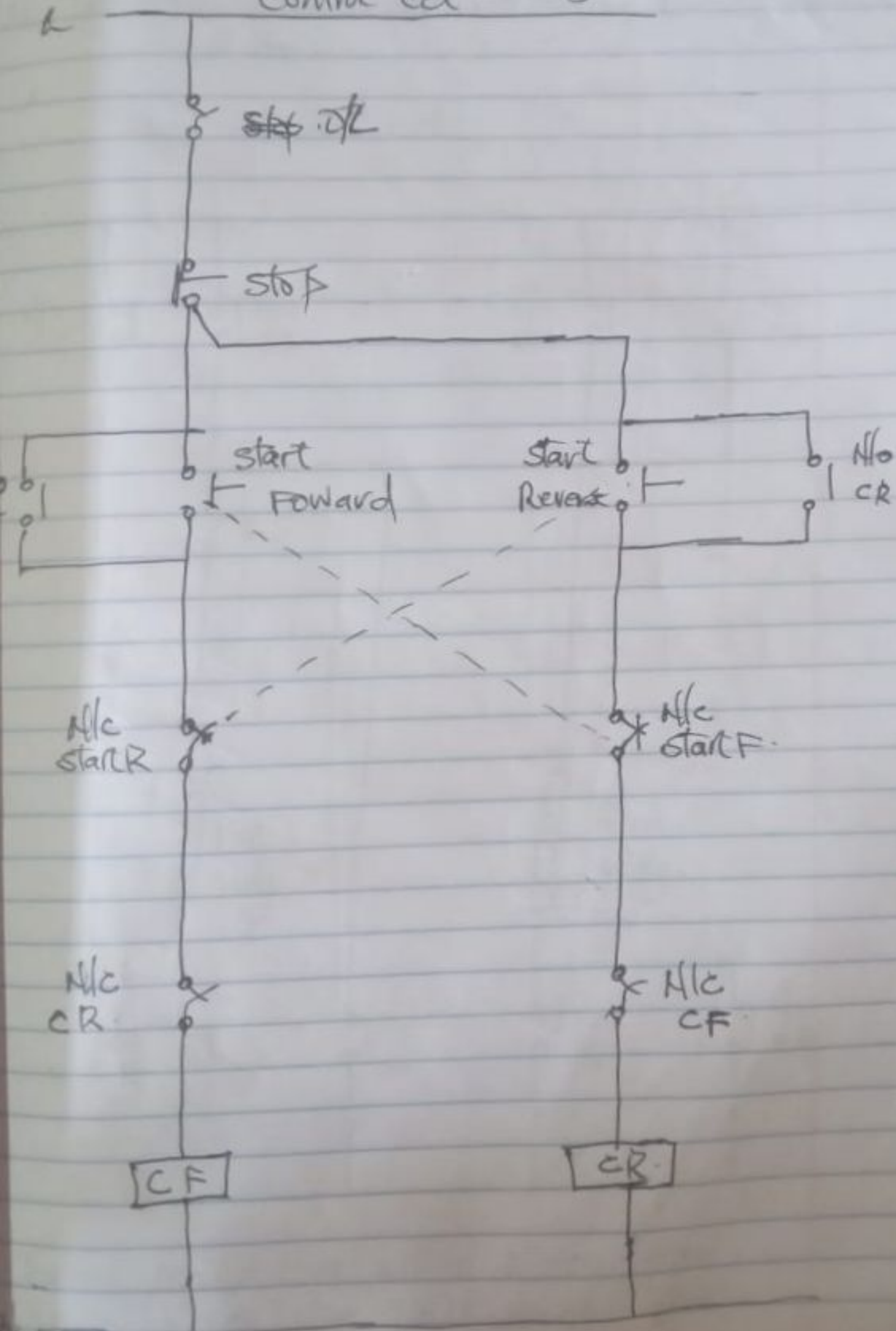


3φ DOL FORWARD REVERSE OPERATION.

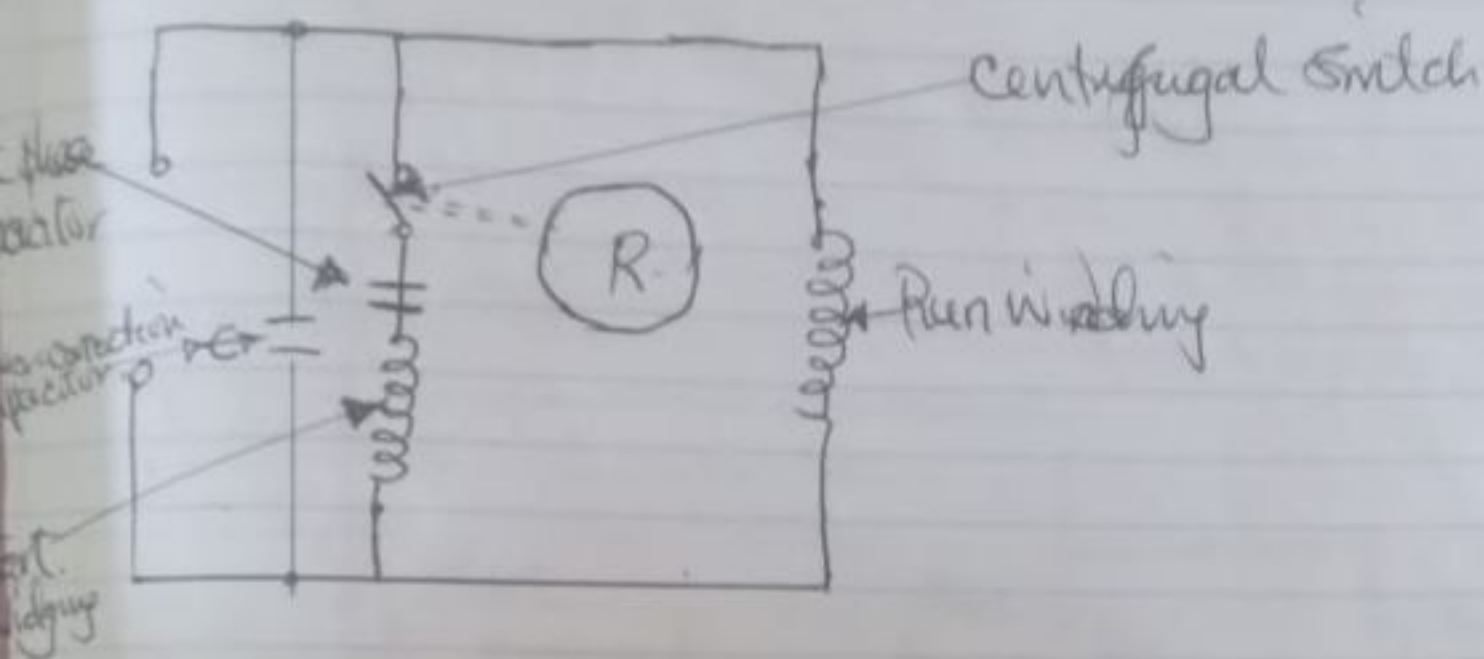
Power Circuit



Ac Induction Motor external wiring Forward-Reverse Run
control ckt.

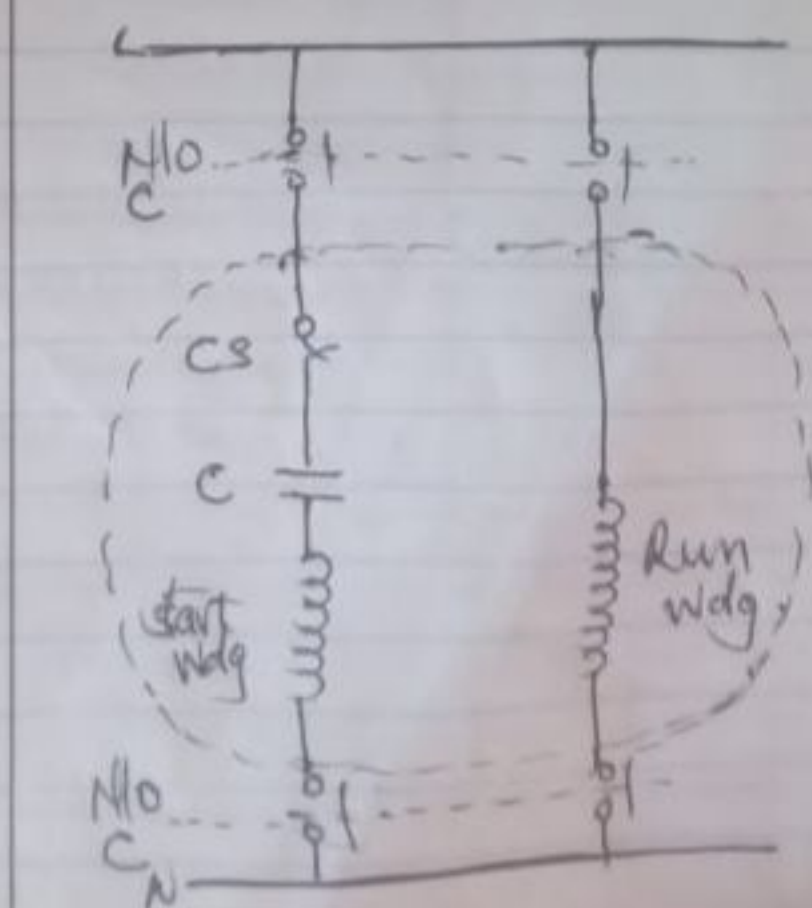
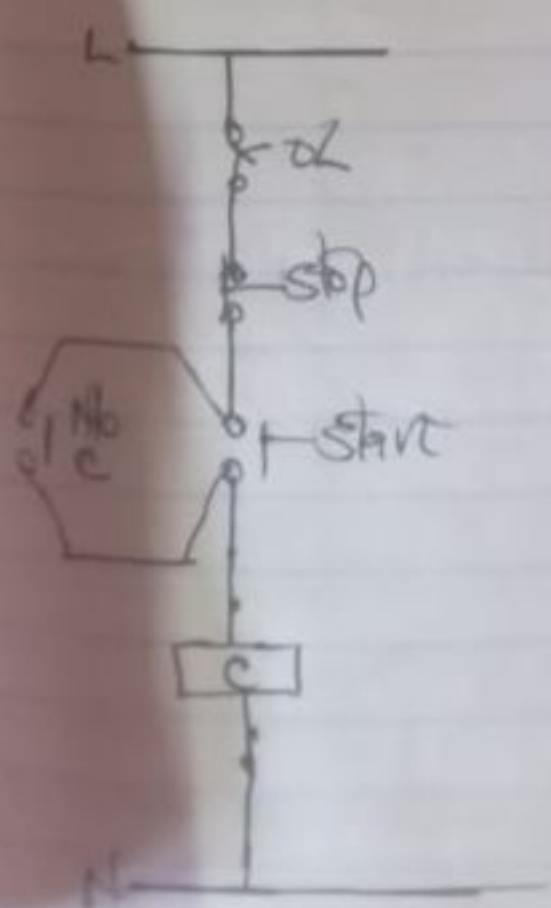


AC Induction motor internal wiring.



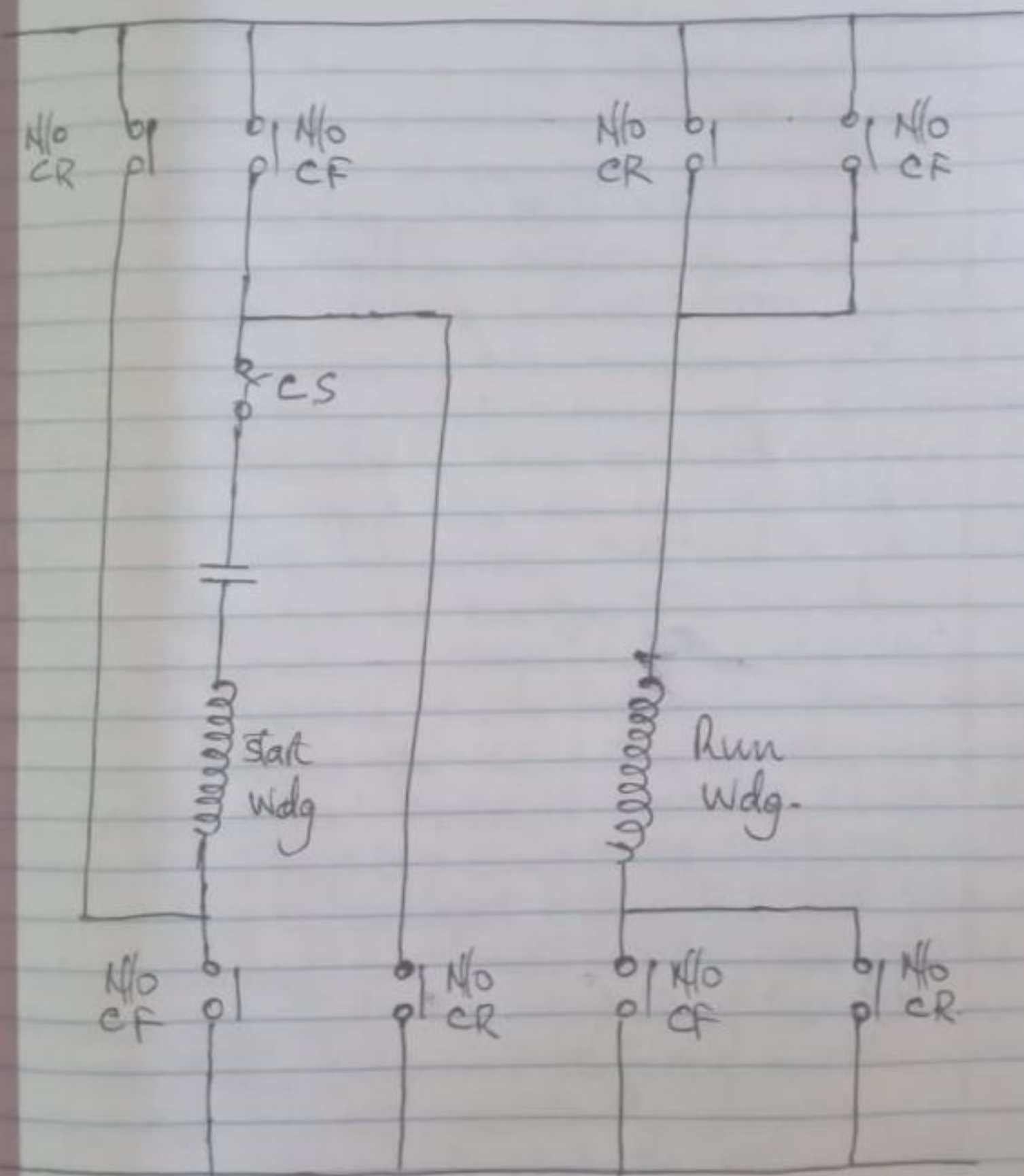
Split phase capacitor connected in series with the start winding to make the start winding more capacitive than the run windings so that the motor create torque for starting.

AC Induction motor external wiring Forward Run.



1 ϕ Induction motor external wiring Forward-Reverse Run

Power cct



Forward N/O CF close while N/O CR open.
Reverse N/O CR close while N/O CF open.